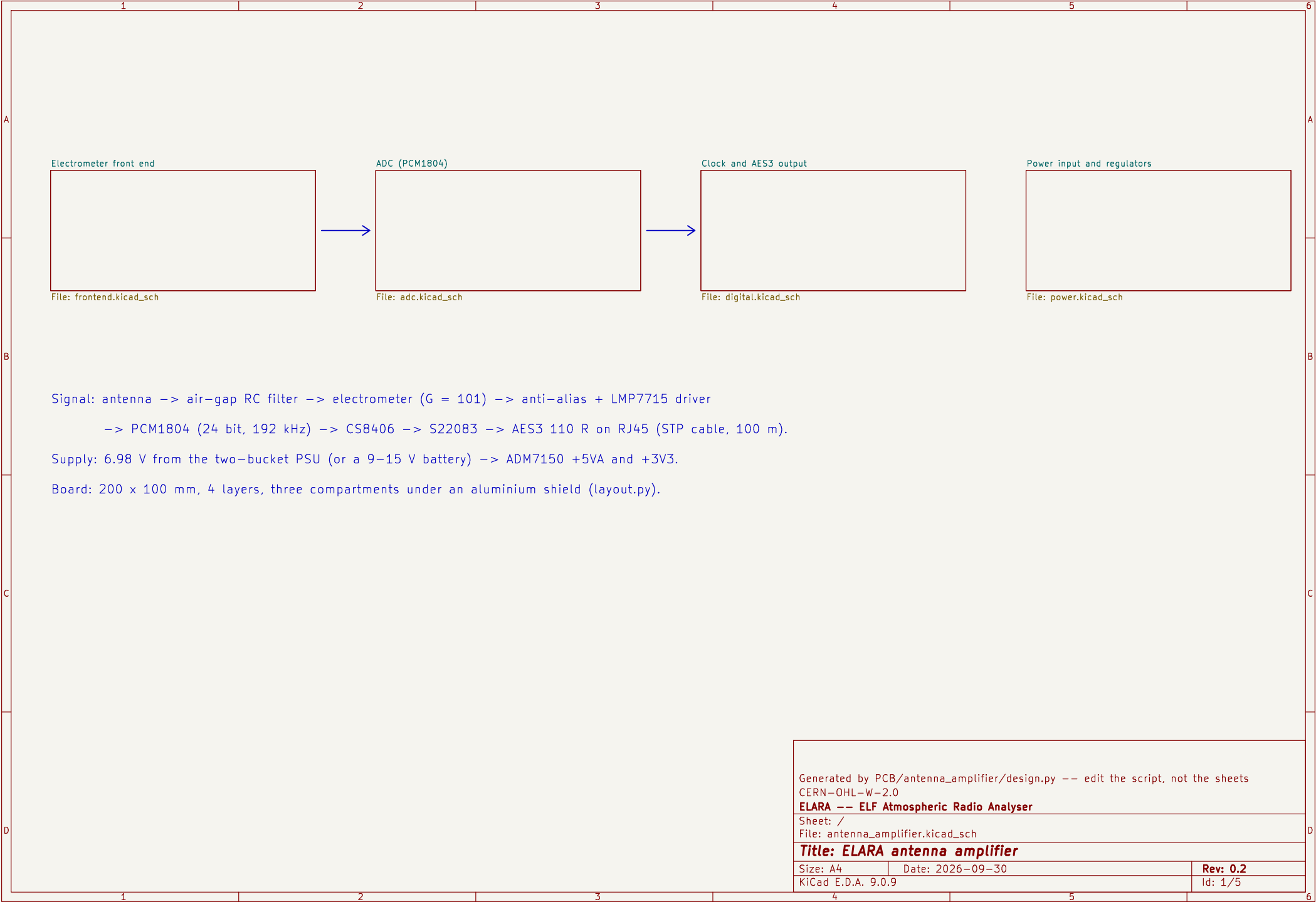




Electrometer front end

ADC (PCM1804)

Clock and AES3 output

Power input and regulators

File: frontend.kicad_sch

File: adc.kicad_sch

File: digital.kicad_sch

File: power.kicad_sch

Signal: antenna -> air-gap RC filter -> electrometer (G = 101) -> anti-alias + LMP7715 driver
-> PCM1804 (24 bit, 192 kHz) -> CS8406 -> S22083 -> AES3 110 R on RJ45 (STP cable, 100 m).
Supply: 6.98 V from the two-bucket PSU (or a 9-15 V battery) -> ADM7150 +5VA and +3V3.
Board: 200 x 100 mm, 4 layers, three compartments under an aluminium shield (layout.py).

Generated by PCB/antenna_amplifier/design.py -- edit the script, not the sheets
CERN-OHL-W-2.0

ELARA -- ELF Atmospheric Radio Analyser

Sheet: /

File: antenna_amplifier.kicad_sch

Title: ELARA antenna amplifier

Size: A4

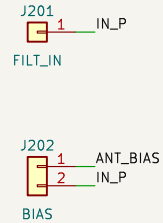
Date: 2026-09-30

Rev: 0.2

KiCad E.D.A. 9.0.9

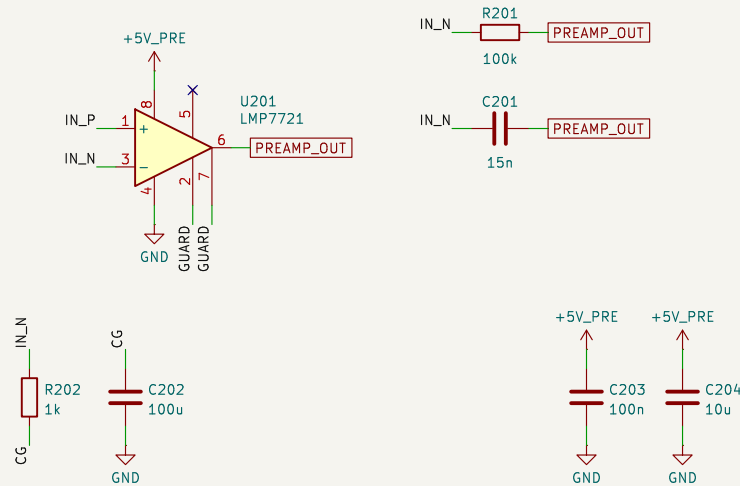
Id: 1/5

INPUT (comp. 1)

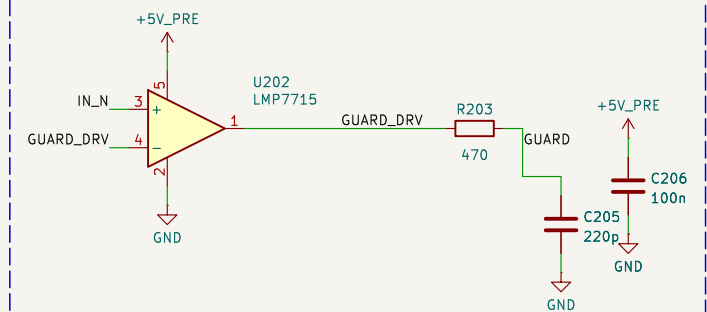


IN_P: guard on all layers,
no mask, PTFE turret.
J202: fit the G-ohm bias R.

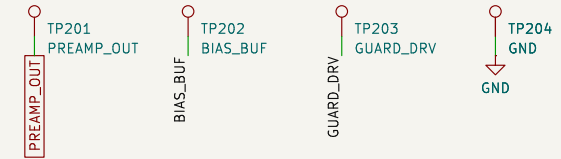
ELECTROMETER $G = 101$



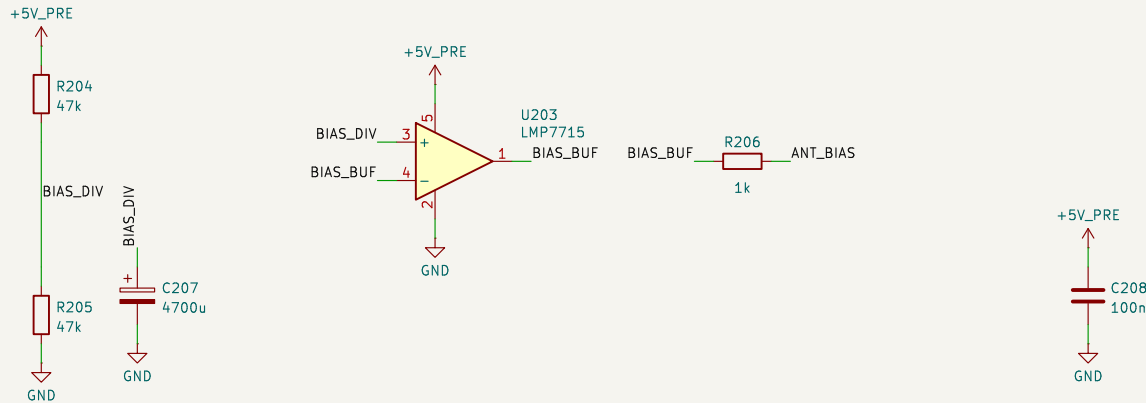
GUARD DRIVER



TEST POINTS



ANTENNA BIAS 2.5 V



Cg returns to GND: the 2.5 V buffer noise stays out of the gain path.
Guard buffer senses IN- (virtual short to IN+): no bias current on IN_P.

Generated by PCB/antenna_amplifier/design.py -- edit the script, not the sheets
CERN-OHL-W-2.0

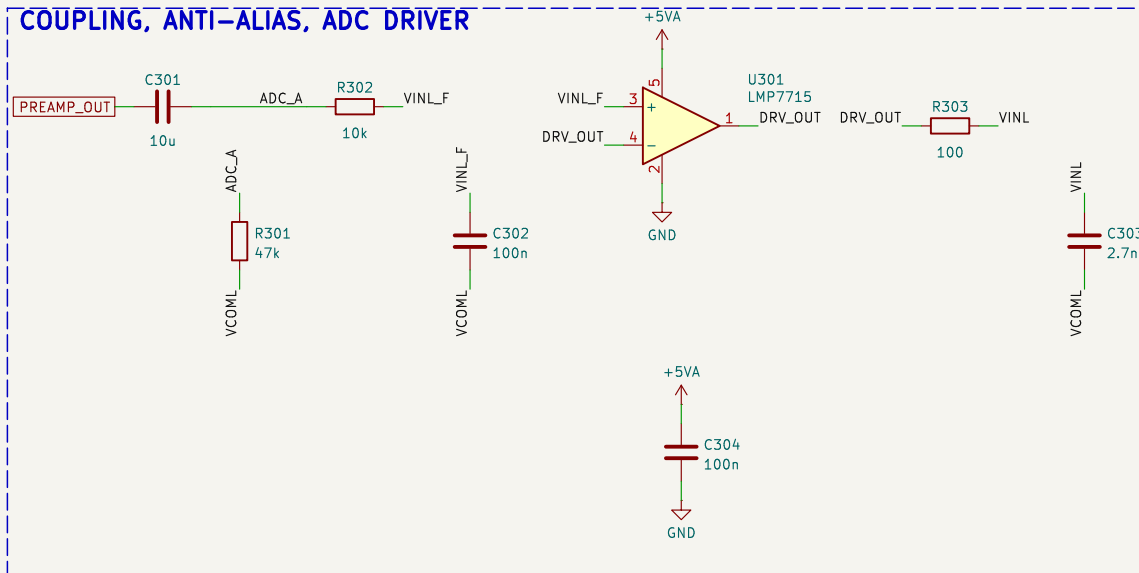
ELARA -- ELF Atmospheric Radio Analyser

Sheet: /Electrometer front end/
File: frontend.kicad_sch

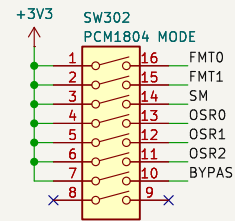
Title: ELARA antenna amplifier -- Electrometer front end

Size: A4 Date: 2026-09-30 Rev: 0.2
KiCad E.D.A. 9.0.9 Id: 2/5

COUPLING, ANTI-ALIAS, ADC DRIVER

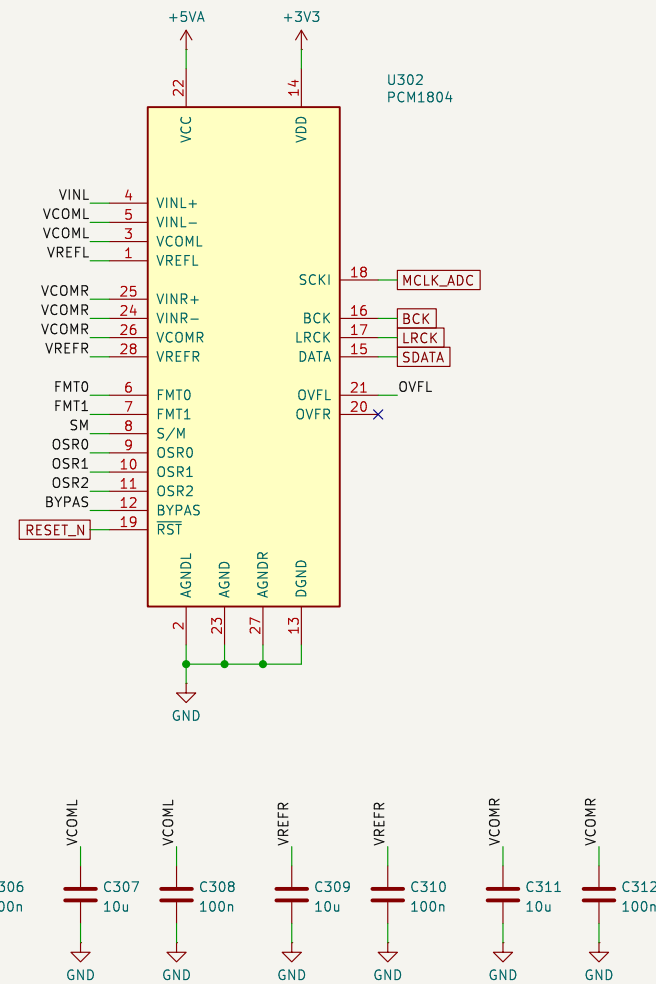


PCM1804 MODE (ON = 1)



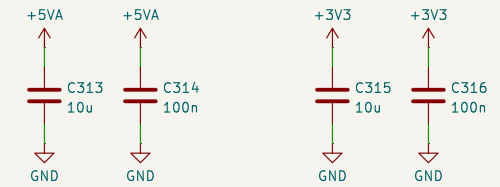
FMT0..OSR2: internal 51 k pull-downs.

PCM1804 master, 192 kHz



Default (192 kHz): S/M=1 master, OSR2..0=111 (SCKI=128 fs), FMT1..0=01 I2S, BYPAS=1 (HPF off).
Change switches with power off, or press RESET afterwards.

SUPPLY DECOUPLING

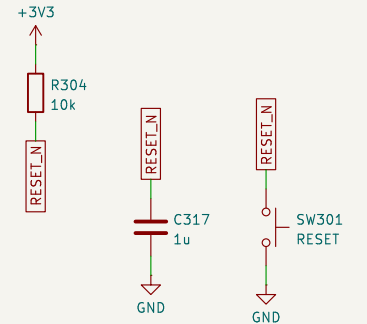


OVERLOAD FLAG

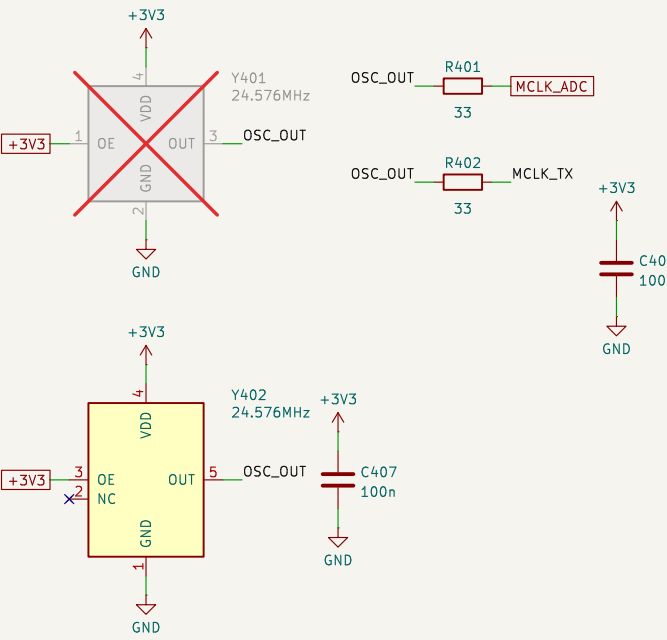


No LED (battery use):
meter or scope on TP301.

RESET

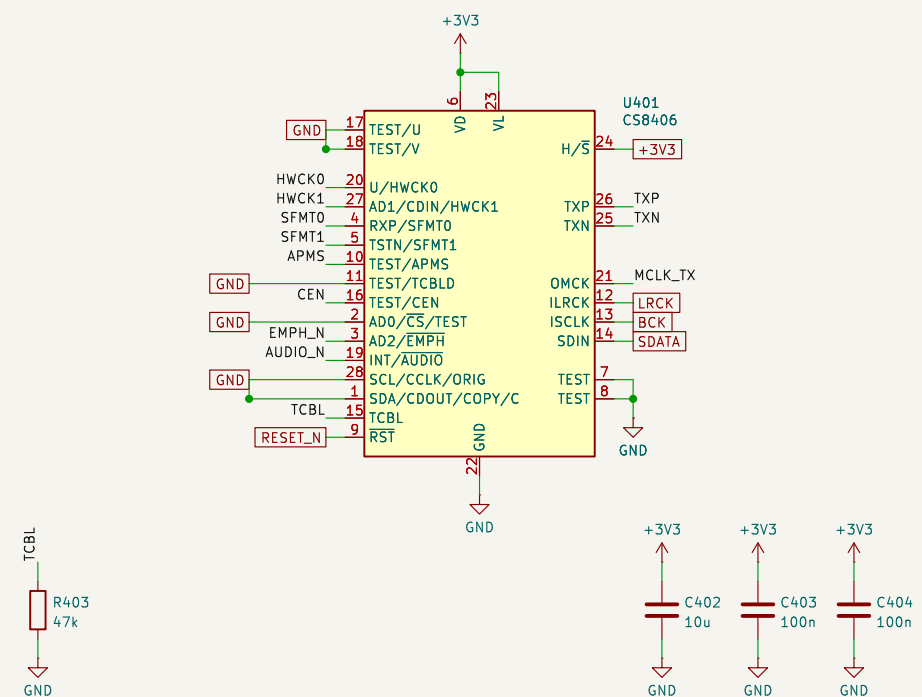


MASTER CLOCK 24.576 MHz

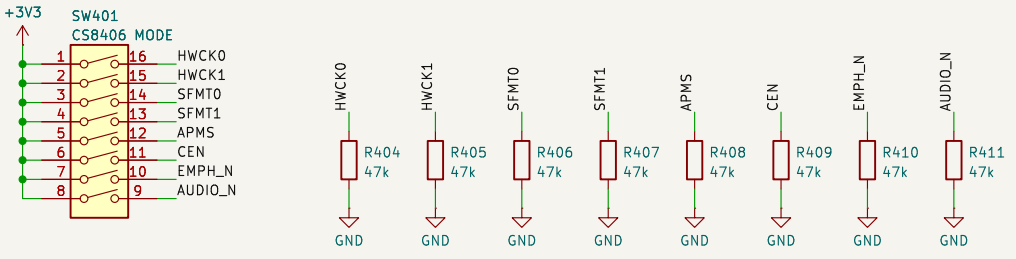


Fit Y402 (SOT23-5, leaded) or Y401 (3.2x2.5 no-lead), never both.

CS8406 hardware mode



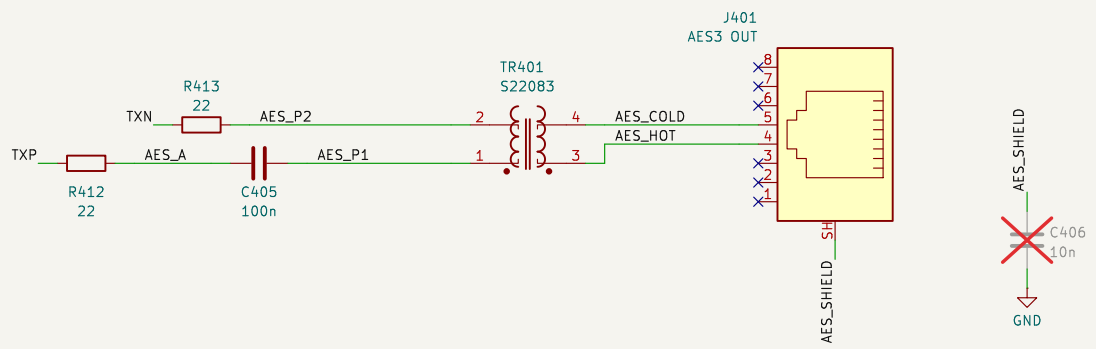
CS8406 MODE (ON = 1)



Default (192 kHz): HWCK -> OMCK = 128 fs, SFMT = 12S, APMS=0 (slave), CEN=0, EMPH_N=1, AUDIO_N=0.

AES3 OUTPUT transformer isolated, RJ45 for shielded twisted pair

AES3 110 R balanced



AES3: $2 \times 22 \text{ R} + 2 \times 33.5 \text{ R driver (CS8406 at VL = 3.3 V)} = 111 \text{ R source}$, 3.3 Vpp into 110 R.
Cable: shielded Cat5e/6 (100 R pairs), shield connected at the indoor end only. AES3 is polarity-free.

